# Credit Card Fraud Detection
# Notebook 1: Exploratory Data Analysis

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# Agenda

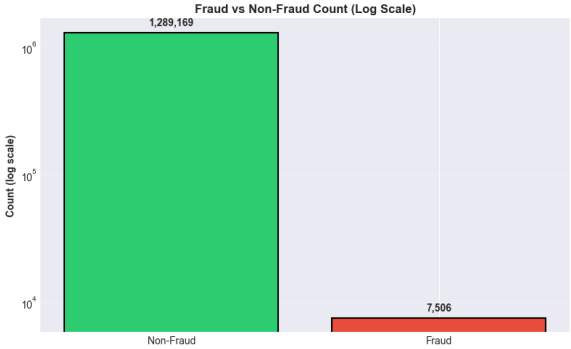
- Dataset overview and quality checks
- Target imbalance analysis
- Geographic and time behavior
- Transaction amount impact
- Other feature signals
- Demo model training and performance
- Final takeaways

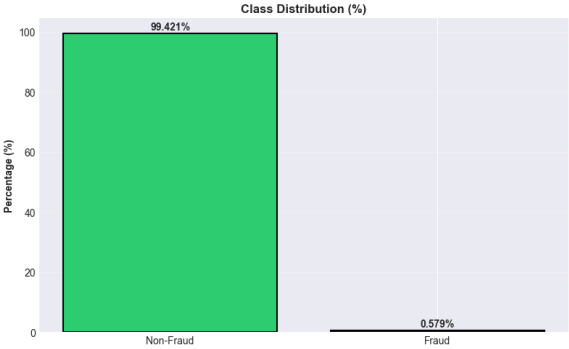
# Dataset Overview

- Training data contains over 1.2M transactions.
- Objective: predict fraudulent transactions in real-world payment activity.
- Target variable: `is_fraud` (0 = non-fraud, 1 = fraud).
- Dropped columns (not used for training): `Unnamed: 0`, `cc_num`, `first`, `last`, `trans_num`.
- Mixed feature groups used in EDA/modeling:
  - Time: `trans_date_trans_time`, `unix_time`, `dob`
  - Amount: `amt`
  - Customer profile: `gender`, `job`
  - Merchant-related: `merchant`, `category`
  - Location: `street`, `city`, `state`, `zip`, `city_pop`
  - Coordinates: `lat`, `long`, `merch_lat`, `merch_long`
  - Engineered feature: `distance_km`

# Target Distribution (Imbalanced Classes)

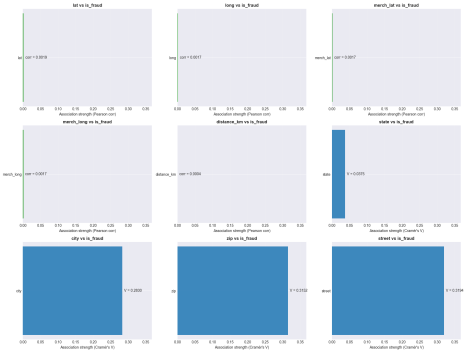
- Fraud class is rare compared with non-fraud.
- Class imbalance must be handled during modeling.



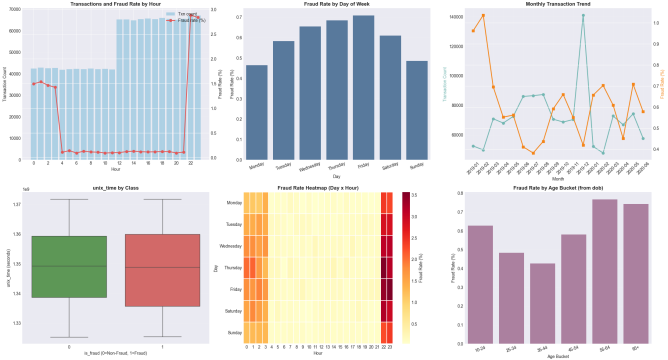
# Geographic Features vs Fraud

- Coordinate-based numeric features have weak association with fraud.
- Location categories (especially `street`, `zip`, `city`) are more informative.



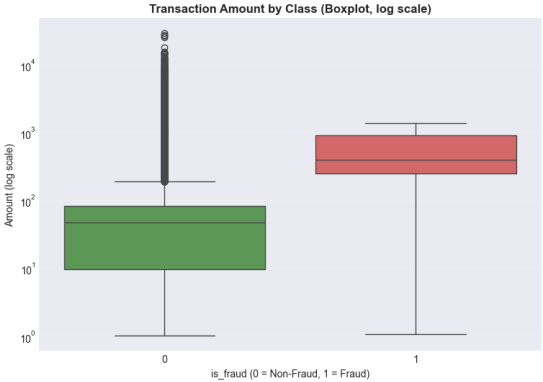
# Time Analysis

- Fraud rate varies by hour and day of week.
- Late-night periods show higher fraud risk.



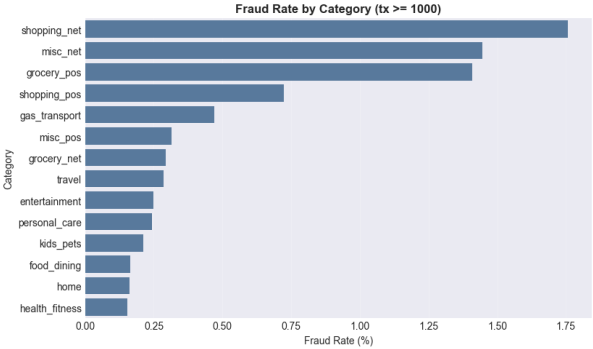
# Transaction Amount Signal

- Fraud transactions are typically higher in amount.
- Boxplot (log scale) shows clear separation between classes.



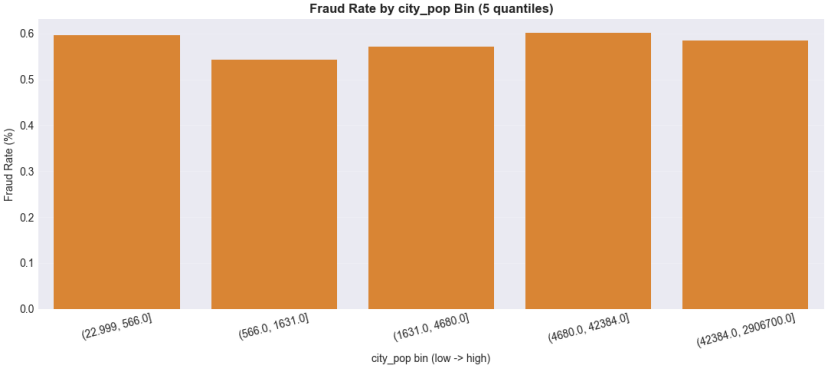
# Other Features I: Gender and Category

- Gender difference exists but is small.
- Categories such as shopping and misc groups show higher fraud rates.



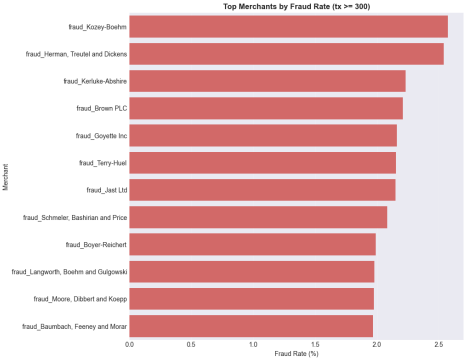
# Other Features II: City Population

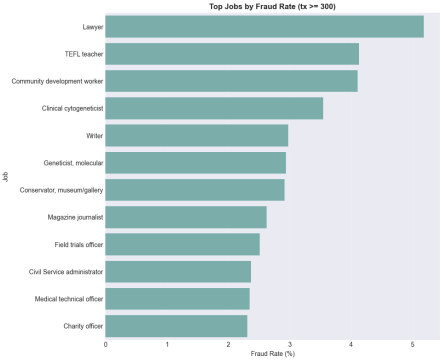
- Fraud rate across city population bins is relatively close.
- `city_pop` alone is weaker than amount/time/category features.



# Other Features III: Merchant and Job

- Some merchant and job groups have noticeably higher fraud rates.
- High-cardinality categorical encoding is important for modeling.



# Demo Model Training

- Source: `https://www.kaggle.com/code/kholoudmohammedzaki/creditcard-model`
- Accuracy: `0.9999745502920933`

## Classification Report

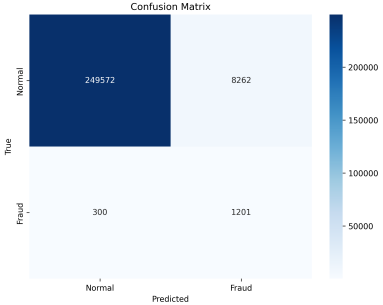
```
Classification Report:

              precision    recall  f1-score   support

           0       1.00      1.00      1.00   1289169
           1       1.00      1.00      1.00      7506

    accuracy                           1.00   1296675
   macro avg       1.00      1.00      1.00   1296675
weighted avg       1.00      1.00      1.00   1296675
```

## Confusion Matrix



# General Conclusion

- The dataset is imbalanced; in future work we plan to handle this with methods such as SMOTE and other imbalance-aware models.
- Some weak-impact features will be tested further and may be removed in future iterations.
- Strong signals include: transaction amount (`amt`), hour (especially late-night), category, and merchant.
- Combining weaker features can still create stronger engineered features and improve prediction power.

# Recommended Next Steps

- Feature engineering: amount transforms, time features, categorical encodings.
- Modeling with class imbalance handling:
  - class weights, focal loss, or resampling
- Evaluate with fraud-focused metrics: PR-AUC, recall, precision.

# Thank you